THE CYCLE CLASS OF A SHEAF FLAT IN HIGH DIMENSIONS IS A RELATIVE CYCLE

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Allowed standard prerequisites (editorial).

- The family of r-cycles associated to a coherent sheaf (relative-cycles chapter, Example example-family-associated-module, canonical tag 0H4Q) and its compatibility with base change (relative-cycles chapter, Lemma lemma-family-associated-module, canonical tag 0H4R). cited at statement line 1116 and proof lines 1126-1127. The object under study and its base-change compatibility, which localizes the specialization check.
- Flatness in dimensions at least n: definition and stability under base change (More on Flatness, Definition flat-definition-flat-dimension-n, canonical tag 0CWG, and Lemma flat-lemma-pre-flat-dimension-n, canonical tag 0CWF). cited at statement line 1119 and proof lines 1124-1125. The hypothesis survives arbitrary base change, needed to apply the valuation-ring reduction.
- Openness of the flat locus for morphisms of finite type (More on Flatness, Lemma flat-lemma-finite-type-flat, canonical tag 05IQ). cited at proof lines 1131-1134. The flat locus U is open, and its complement has fibres of dimension below r, so restriction to U identifies the r-cycle groups compatibly with specializations.
- Specialization of the cycle class of a flat coherent sheaf over a valuation ring (relative-cycles chapter, Lemma lemma-specialization-module, canonical tag 0H4I). cited at proof lines 1142-1143. Settles the reduced flat case, finishing the proof.

Source reference. Original reference 0H4Q: sources/stacks-chow/tags.txt, source lines 21360–21360. Cited at statement line 1116: the situation in which the lemma lives; kind reference as example is outside the assembly whitelist.

Source reference. Original definition 0CWG: sources/stacks-chow/tags.txt, source lines 16171–16171. Cited at statement line 1119: the flatness hypothesis.

Source reference. Original lemma 0CWF: sources/stacks-chow/tags.txt, source lines 16170–16170. Cited at proof lines 1124-1125: the hypothesis is preserved by base change.

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Source reference. Original lemma 0H4R: sources/stacks-chow/tags.txt, source lines 21361–21361. Cited at proof lines 1126-1127: the cycle-class formation commutes with base change.

Source reference. Original lemma 05IQ: sources/stacks-chow/tags.txt, source lines 7139–7139. Cited at proof lines 1131-1134: the flat locus U is open on the valuation ring base.

Source reference. Original lemma 0H4I: sources/stacks-chow/tags.txt, source lines 21353–21353. Cited at proof lines 1142-1143: settles the flat case, finishing the proof.

Lemma 0.2. *In the situation of Example 0H4Q assume S is locally Noetherian and $\mathcal{F}$ is flat over S in dimensions $\geq r$ (More on Flatness, Definition 0CWG). Then $[\mathcal{F}/X/S]_r$ is a relative r-cycle on X/S.*

Proof. By More on Flatness, Lemma 0CWF the hypothesis on $\mathcal{F}$ is preserved by any base change. Also, formation of $[\mathcal{F}/X/S]_r$ is compatible with any base change by Lemma 0H4R. Since the condition of being compatible with specializations is checked after base change to the spectrum of a discrete valuation ring, this reduces us to the case where S is the spectrum of a valuation ring. In this case the set $U = \{x \in X \mid \mathcal{F} \text{ flat at } x \text{ over } S\}$ is open in X by More on Flatness, Lemma 05IQ. Since the complement of U in X has fibres of dimension $< r$ over S by assumption, we see that restriction along the inclusion $U \subset X$ induces an isomorphism on the groups of r-cycles on fibres after any base change, compatible with specialization maps and with formation of the relative cycle associated to $\mathcal{F}$. Thus it suffices to show compatibility with specializations for $[\mathcal{F}|_U/U/S]_r$. Since $\mathcal{F}|_U$ is flat over S, this follows from Lemma 0H4I and the definitions. □